

Phase 4: Develop

Presentation Guidelines

- 15 Minutes for the presentation
- 5 Mins for Q&A

Introduction Slides (2-3 minutes)

Title Slide

Must contain:

- ☐ Project Title
- ☐ Sponsor company name & logo
- ☐ Team number
- ☐ "Phase 4"
- ☐ "DAEN 460 - Capstone"
- ☐ A Texas A&M logo

Team Introduction Slide 1

- ☐ Pictures & names for each group member
- ☐ Phase 4 Fun Fact: *What is a hobby/skill/interest you hope to pick up after you've graduated?*

Team Introduction Slide 2

- ☐ Picture and name of project sponsor
 - You can use the sponsor's logo instead of a headshot/photo if there's not an openly accessible picture of them online.
- ☐ Picture and name of faculty advisor

Team Introduction Slide 3

- ☐ Problem statement slides from presentation 1 (as revised).

Project Recap

- ☐ Take 1-2 minutes to concisely give an overview of your project. Focus on delivering a high-level overview that will prepare the audience to understand how and why you are going forward with your final design. Ideally, the overview will be 1-3 slides, consisting almost entirely of visuals.

Body Slides (7-9 minutes)

1. Final Design Architecture

- **Objective:** Provide a top-down explanation of your final design, ensuring that both technical and non-technical stakeholders can understand the major components and their interactions. (See additional tips below if your project is research focused)
 - **Top-Down Approach:** Start with a high-level view of the system, outlining the key components or subsystems and their roles. Break down the interactions between these components. Each component's purpose should be made clear before diving into any details. This structure helps your audience understand how each part contributes to the overall solution.
 - **Focus on Functionality:** Clearly define what your system is intended to accomplish and how it achieves that goal. Avoid getting into technical minutiae unless it is essential to show how the design meets a critical requirement.
- **Mapping Requirements to Design Features:** Instead of showing the entire traceability matrix (which may be too detailed for a presentation), focus on critical requirements and how they are satisfied by the design. You can pull key requirements from Phase 2 and showcase how the final architecture satisfies each.
 - Use specific examples, like how a new algorithm reduces cycle time or how a redesigned workflow improves throughput.
 - **Customer Value Focus:** Always return to why your solution is valuable from the customer's perspective. Highlight the improvements your design makes over the existing system (the "As-Is" state). If your solution reduces costs, improves efficiency, or mitigates risks, make these points clear in this section.
- **Salient Results:** Each visual should imply something significant. For example, a graph that shows efficiency improvements or cost savings should have clear annotations pointing out the most important takeaways. Avoid cluttering your visuals with unnecessary details; focus on what you want the audience to remember.

Additional Tips for presenting the Final Design Architecture:

- **Keep the Narrative Flowing:** Ensure that your explanation of the design architecture is logical and flows naturally. Avoid jumping back and forth between unrelated components; instead, build a narrative that progressively introduces each part of the design.
- **Highlight Improvements:** Where possible, quantify the benefits of your final design. Did it reduce costs by 15%? Improve throughput by 30%? Reduce error rates to near-zero levels? These quantifiable results will help make a compelling case for your design.

- **Prepare for Q&A:** Since you won't present the entire traceability matrix, be ready to reference specific points if asked during the Q&A. Keep supporting documents in your appendix to back up your claims.
- **Research Projects:** Describe the areas of research that you studied and briefly summarize the results. Map the results of your research to your requirements and explain how the requirements are met.

2. Analysis: Economic, Performance, Risk +++

- In Phase 3, some analyses were conducted to compare alternatives. Now, you will be analyzing your selected solution through relevant lenses of analysis.
- Your analysis shouldn't be limited to economic, performance, and risk. While these are generally the big three of systems analysis, it is up to your group to find additional means of evaluating the behavior of your final design.
- For research, discuss the impact of the research and any analysis regarding validity, implementation, or other salient factors.
- **Even if some groups may find it challenging to quantify the economic, performance, and risk dimensions, it is essential to make a diligent effort to at least estimate the impact of the system or project solution on the system it is being implemented into.**
 - Quantifying these impacts, even through estimates, provides essential information about the overall value and feasibility of the solution.

Think of what kind of information your sponsor's boss would like to see! Then, find ways to verifiably quantify and visualize those key points.

3. Engineering Standards and Project Constraints

- **Engineering Standards:** Elaborate on the relevant engineering standards that guided the technical aspects of your project. These might include industry-specific standards, safety regulations, or best practices that are widely recognized in the field of industrial and systems engineering. Some examples might be:
 - ISO Standards (International Organization for Standardization) relevant to the design or quality management processes you followed.
 - OSHA (Occupational Safety and Health Administration) regulations, if safety was a concern in your project.
 - ASTM Standards (American Society for Testing and Materials) for any testing protocols you might have used.
 - HIPAA (Health Insurance Portability and Accountability Act), if your project dealt with sensitive healthcare data.
 - NIOSH (National Institute for Occupational Safety and Health) guidelines for ergonomic considerations.

- **Project Constraints:** Discuss the various constraints your team faced throughout the project. These could include:
 - **Technical Constraints:** Limitations in technology, tools, or resources that influenced your design choices.
 - **Time Constraints:** Any time limitations, such as deadlines imposed by the sponsor or the course schedule, that affected your project timeline.
 - **Economic Constraints:** Budget limitations that influenced the design and implementation of your project.
 - **Legal and Regulatory Constraints:** Any legal issues or regulatory requirements that had to be followed, such as environmental regulations or data privacy laws.
 - **Ethical Constraints:** Constraints related to ethical considerations, such as ensuring that the final solution does not negatively impact stakeholders or violate professional ethics.

Conclusions & Recommendations (3-4 minutes)

Wrap up your presentation by giving a conclusion to your semester's efforts. Recall the concept of "the narrative" brought up in preparation for Phase 3. This is the time to recap the story of your Capstone project.

Going hand in hand with conclusions, you should also cover any key recommendations you will be making to your sponsor. These recommendations can come in many forms, two popular methods described below:

- **Implementation Plan**
 - Outline the steps required to bring the final design to reality, including timelines and resources needed
 - Elaborate on how you plan to mitigate any concerns that arose in your cost, performance, risk, or other analyses
 - Your team may not have created something that's going "into production". If that's the case, your implementation plan will be focused on how the sponsor can make use of your final design/findings.
 - **Think back to why your project was set in motion in the first place!** How will your semester's work be implemented?
- **Test plan**
 - Describe the testing procedures to validate the performance and reliability of the final design
 - Testing may be possible before your group "hands off" your solution to your sponsor, or not. Either way, your group should come up with some set of executable steps to help verify the functionality of your solution.

- If your project is more research-based, then it may be hard to "test" your final design. In this case, provide some elaboration on the limitations of your final findings. Then, describe how additional research efforts may redirect or improve your conclusions.
- If your sponsor has indicated to you that your project could potentially be extended to another Capstone team in a future semester, then you should create a "package" of documentation, models, artifacts, etc. and compile any tips or instructions to support the future team(s).

Appendix

- Your appendix should include any additional documentation, figures, models, or data that may aid you in answering questions following your Q&A.
- The addition of your supplemental work in your appendix will also make it easier for the instructors to give you credit for your technical work.
 - **We only see what you include in your slide deck!**
- It helps to organize your appendix, a table of contents can be beneficial
- **IMPORTANT:** For any artifacts created that follow a methodology/style you found from an online resource, you must include your source(s) in the respective deliverable submission and the appendix.

ABET Accreditation Alignment

To ensure your Capstone project meets the professional standards required for ABET accreditation, **your Phase 4 presentation and final report must explicitly demonstrate your ability to apply engineering design to produce solutions that meet specified needs.** While these concepts are woven throughout your slides, use the following checklist to ensure you have explicitly addressed the required Student Outcomes:

Public Health, Safety, and Welfare

In your "Engineering Standards and Project Constraints" slide, go beyond just naming a standard (like OSHA or ISO). Clearly state how your design specifically protects the user or the public. If your design is implemented, what are the safety implications? How did you mitigate risks to human welfare?

Global, Cultural, Social, and Environmental Factors

In your "Analysis" section, consider the broader impact of your design. Global/Cultural: Does your solution account for diverse user bases or different geographical regulations? How does this improve the working conditions or lives of the stakeholders? Does your design minimize waste,

energy consumption, or carbon footprint? **Even for digital or research-based projects, discuss the "sustainability" of the process or implementation.**

Professional and Ethical Responsibility

Ensure your narrative reflects informed, ethical decision-making. **Address any ethical dilemmas encountered.** This includes data privacy (HIPAA/GDPR), ensuring unbiased algorithms, or being transparent about the limitations of your research.

Consideration of Engineering Standards

This is a mandatory ABET requirement. You must explicitly identify the specific standards (e.g., IEEE, ASME, NIST, ISO) that informed your design. If no specific industry standard exists, you must describe the "best practices" or internal corporate standards provided by your sponsor that served as your technical framework. Be prepared to explain why a specific constraint (like budget or a legal regulation) forced a design change. Evaluators look for the link between a real-world constraint and your final engineering "trade-off" decisions.